POLITECNICO MILANO 1863 Academic year 2023/2024	Course RF SYSTEMS	**Person code (codice Persona)** _____
	Student Signature _____	**Student Name, Surname** _____

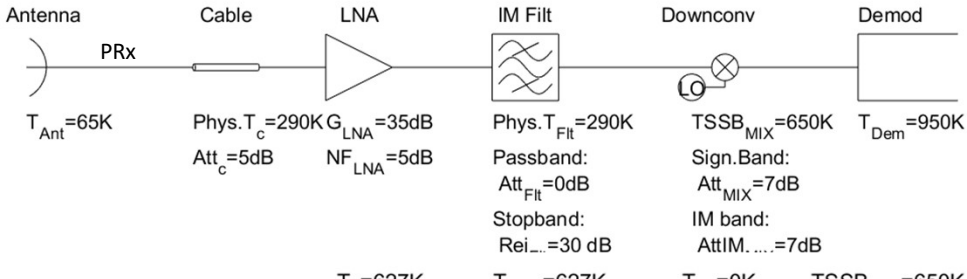
11 July 2024 Exam (27 pts available; ≥15 needed)

Please answer the following questions/problems, providing a brief explanation in English language of the steps/computations involved. Please specify units for all numeric results requiring them. Allowed support material: textbooks, notes, scientific calculator, laptop/tablet (offline). Write with a pen.

Exercise 1 (10 points)

A receiver operating at 18GHz receives a 26MHz-wide signal centered around the carrier with a power of PRx=-72dBm and downconverts it to an intermediate frequency of f_{IF}=250MHz.

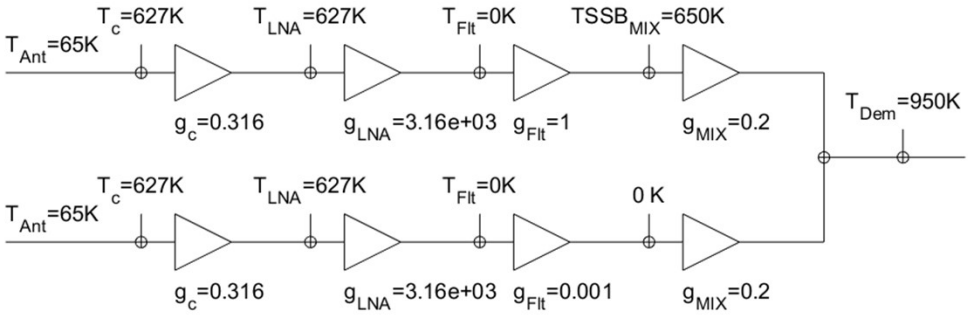
a) Draw the equivalent noise diagram for the receiver architecture shown below and compute the system noise temperature (at the antenna port).
b) What is the equivalent noise power in the 26MHz band and SNR at the antenna port? What is the equivalent noise power in the 26MHz band and SNR at the demodulator input?
c) Choose a suitable frequency for the local oscillator and determine the corresponding image band specifying its minimum and maximum frequencies.
d) An interferer exactly in the image band c) is received as a signal of -60dBm at the antenna port of our receiver. Modeling this interfering signal as additional thermal white noise, what is the achievable data rate to guarantee Eb/N0=12dB? (The IM filter attenuates the interferer!)



Answers

a) All gains are converted to lin. scale. NFs are converted to equivalent input noise temp: $Teq=T_0(NF-1)$. The cable has NF equal to its attenuation and instead uses the physical temperature

$T_c=T_{phys}*(Atten-1)$



The image band is only partially suppressed by the IM filt, due its limited stopband rejection, thus image-band contributions, such as the wideband noise from antenna, cable, LNA will reach the IF, although attenuated.

Tsys=Tant+Tc+TLNA/gc+Tflt/gc/gLNA+TSSBMix/gc/gLNA+Tdem/gc/gLNA/gMix+
((Tant+Tc)*gc*gLNA*gFltIM*gMix+Tlna*gLna*gFltIM*g<ix) /gc/gLNA/gMix =2683K

b) Pneq@AntennaPort=k*Tsys*B=0.962pW (=-90.2dBm); Eq. SNR@antenna port=-72dBm+90.2dBm=18.2dB
Pneq@DemInput= Pneq@AntennaPort–Attc+GLNA-AttMix=-67.2dBm; Signal@DemInput=PRx–Attc+GLNA-AttMix = -49dBm → Eq. SNR@DemInput=-49dBm+67.2dBm=18.2dB (In fact,SNR with equiv. noise is constant in the chain!)

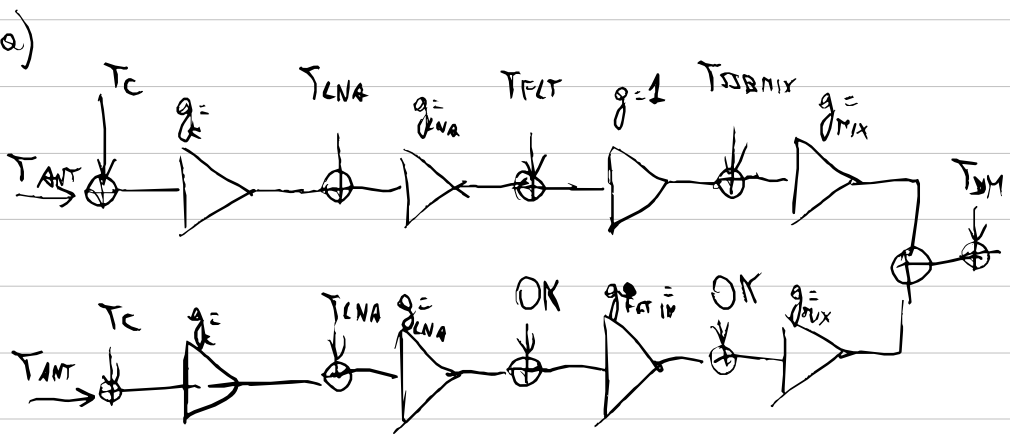
c) A reasonable choice for the LO is fLO =18GHz-250MHz=17.75GHz (the other would be 18.25GHz). With fLO=17.75GHz, the image band occurs where |fIM±fLO|=fIF→the image band occurs around fIM=17.5GHz and has bandwidth equal to the useful signal, that is 26MHz. Thus the image band spans from 17.487GHz to 17.513GHz.

d) At the dem. (easier): $P_{int@Dem}=P_{int}-Ac+G_{LNA}-Rej-A_{Mix}$=-67dBm. Tot undesired@Dem=$Pn_{@Dem}+P_{int@dem}$=0.39nW (=-64.1dBm). Tot SNIR@Dem=-49dBm-(-64.1dBm)=15.1dB. Alternatively: the total equivalent undesired signal @antenna port is $Pn_{tot}=Pn_{eq}+P_{int}$=0.926pW+1pW=1.926pW(=-87.1dBm). Therefore,SNIR@Ant=-72+87.1=15.1dB. Anyway, SNR =(Eb/N0)*R/B → R=B*SNR/(Eb/N0)=53.1Mbit/s

EX1 11/07/24 10:41 → 11:11 · 30 MIN

$f = 18\ GHz$ $BW = 26\ MHz$ $P_{RX} = -72\ dBm$ $f_{IF} = 2.5\ GHz$

a)



$$T_C = T_{ANT}\left(10^{\frac{5}{10}} - 1\right) = 627\ K$$

$$g_c = 10^{-\frac{5}{10}} = 0.316$$

$$T_{LNA} = T_0\left(10^{\frac{5}{10}} - 1\right) = 627\ K$$

$$g_{LNA} = 316.2$$

$$T_{FLT} = 0\ K$$

$$g_{FLT\,IM} = 10^{-\frac{30}{10}} = 0.001$$

$$g_{MIX} = 10^{-\frac{7}{10}} = 0.2$$

$$T_{SYS} = T_{ANT} + T_C + \frac{T_{LNA}}{g_c} + \frac{T_{FLT}}{g_c g_{LNA}} + \frac{T_{SSB}}{g_c g_{LNA}} + \frac{T_{DSM}}{g_{MIX} g_{LNA} g_c}$$

$$+ \left(\left(T_{ANT} + T_C\right) g_c g_{LNA} g_{FLT\,IM} \cdot g_{MIX}\right) \frac{1}{g_{MIX} g_{LNA} g_c}$$

$$+ \left(T_{LNA}\, g_{LNA}\, g_{FLT\,IM} \cdot g_{MIX}\right) \frac{1}{g_{MIX}\, g_{LNA}\, g_c}$$

$$= \left(T_{ANT} + T_C\right)\left(1 + g_{FLT\,IM}\right) + \frac{T_{LNA}\left(1 + g_{FLT\,IM}\right)}{g_c} + \frac{T_{FLT}}{g_c g_{LNA}} + \frac{T_{SSB}}{g_c g_{LNA} g_{MIX} g_c} + \frac{T_{DSM}}{g_{MIX} g_{LNA} g_c}$$

$$= 2684\ K$$

b) $P_{noise@ANT} = KT_{SYS} \cdot B = 0.963\ pW \quad (= -90.2\ dBm)$

$$SNR^{SG}_{@ANT} = -72\ dBm - (-90.2)\ dBm = 18.2\ dB$$

$$P_{noise@DSM} = P_{noise@ANT} \cdot g_c \cdot g_{LNA} \cdot g_{MUX}$$

$$= P_{noise@ANT} - A_C + G_{LNA} - A_{MUX} = -67.2\ dBm$$

SNR^{SG} REMAINS CONSTANT ACROSS THE CHAIN

$$P_{sig@DSM} = P_{RX} - A_C + G_{LNA} - A_{MUX} = -49\ dBm$$

$$SNR^{SG}_{@DSM} = P_{SIG@DSM} - P_{noise@DSM} = SNR^{SG}_{@ANT} = 18.2\ dB$$

c) $f_{IF} = f_{RF} - f_{LO}$

$$\Rightarrow f_{LO} = f_{RF} - f_{IF} = 17.75\ GHz$$

The IMAGE BAND is 26 MHz wide and centered at $f_{LO} - f_{IF} = 17.5\ GHz$

$$\Rightarrow \text{IMAGE BAND: } 17.5\ GHz - 13\ MHz,\ 17.5\ GHz + 13\ MHz$$

$$[17.487\ GHz,\ 17.513\ GHz]$$

d) $P_{INTERF} = -60\ dBm$ in IMAGE BAND

At the DEMOD INPUT:

$$P^{@DSM}_{INTERF} = P_{INTERF} - A_C + G_{LNA} - REJ_{FLT} - A_{MUX} = -67.0\ dBm$$

$$P^{@DSM}_{UNDESIRED} = P^{@DSM}_{INTERF} + P^{@DSM}_{n} = 10^{-\frac{67}{10}}\ mW + 10^{-\frac{67.2}{10}}\ mW = 0.39\ mW = -64.1\ dBm$$

$$\Rightarrow SNR^{@DSM}_{TOT} = -49\ dBm - (-64.0\ dBm) = 15.1\ dB$$

$$\Rightarrow SNR = \frac{P_S}{P_N} = \frac{E_B / T_B}{N_0 \cdot B} = \frac{E_B \cdot R}{N_0 \cdot B}$$

$$\Rightarrow \frac{R}{B} = \frac{10^{\frac{15.1}{10}}}{10^{\frac{12}{10}}} \cdot 2.04 \Rightarrow R = 53.1\ Mbit/s$$